# Scope Dependence of Strict Splitting Counts
## An Exact Check of 69 Recorded Seed Positions

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**Abstract**

We test the previously reported lower strict-splitting-count criterion on a broader, previously certified census of 69 prime-order seed positions, retaining all 82 recorded root branches. No cutoff from 100 through 5000 separates the ten published positions from all 59 unpublished positions. At the original cutoff 691, 22 unpublished positions tie or fall below their field's published maximum. They belong to categories excluded from the earlier 21-choice cohort by additional root and order-width screens. A separate PARI implementation reproves 76 distinct field/order primes, verifies the point certificates and reconstructs all six output files; three deliberate corruptions are refused. The earlier 21-choice result reproduces exactly. This scope test limits the splitting criterion alone; it neither refutes a combined-screen hypothesis nor establishes historical selection, random-curve unusualness or an ECDLP weakness.

# 1 Fixed criterion and population

For a prime-field curve of certified full prime order $n$, define

$$t = p + 1 - n, \quad \Delta = t^2 - 4p, \quad S_L = \#\{\text{odd primes } \ell \le L : (\Delta/\ell) = +1\}.$$

Let $R_L$ count the separate zero-character cases. A strict split prime has two distinct Frobenius-stable lines in $E[\ell]$, corresponding to cyclic isogeny kernels defined over the base field. Their individual points need not be rational. Ramified cases can also admit rational isogenies; $S_L$ therefore does not count every rational small-degree isogeny [3, 5].

The degree-691 observation predates this analysis [1]. The earlier 21-choice replay compares ten publications with eleven named unpublished candidates. Their lower-count separation survives only at integer cutoffs 691–718 and 769–772. The inherited November-1997-draft acceptance analysis records a broader prime-order cohort: 69 positions, of which ten are published and 59 unpublished [2]. Its additional 48 positions comprise 17 width-only and 31 other-root-only cases. This is a lower-bound census of numerically eligible recorded positions, not evidence of original visits, rejections or an exhaustive near-prime search.

The fixed comparison is the same: within each field, all unpublished counts must exceed the largest published count. At each integer $100 \le L \le 5000$, the margin is $m_L = \min_{\text{unpub}} S_L - \max_{\text{pub}} S_L$; strict success requires $m_L > 0$. The broader census has five paired fields. P-521 remains a singleton and receives no artificial margin. All labels were visible; this is retrospective reanalysis, with no fitted replacement cutoff.

# 2 Broader result

| Bits | Pub. | Unpub. | Pub. $S_{691}$ | Unpub. $S_{691}$ | Margin | Pos. cutoffs | Violators |
|---|---|---|---|---|---|---|---|
| 192 | 3 | 23 | 57–60 | 44–71 | -16 | 0/4901 | 14 |
| 224 | 1 | 11 | 50–50 | 50–66 | 0 | 1801/4901 | 1 |
| 239 | 3 | 14 | 42–57 | 54–71 | -3 | 0/4901 | 5 |
| 256 | 1 | 5 | 57–57 | 56–67 | -1 | 0/4901 | 2 |
| 384 | 1 | 6 | 54–54 | 56–60 | 2 | 635/4901 | 0 |
| 521 | 1 | 0 | 52–52 | — | — | — | — |

The ranges refer to $S_{691}$. Positive-cutoff counts cover all 4901 integers. "Violators" are unpublished positions with $S_{691}$ at or below the published maximum in the same field; equality fails the strict rule. No broader joint strict or inclusive $(S_L + R_L)$ separator occurs anywhere in the requested range. P-192, P-239 and P-256 each fail individually at every cutoff. This does not change the correctly reproduced narrow result.

## 3 Counterexamples and the role of additional screens

| Recorded category | Positions | Violators at691 |
|---|---|---|
| Other-root-only | 31 | 14 |
| Order-width-only | 17 | 8 |
| Named omissions | 11 | 0 |

All 22 counterexamples lie among the extra 48 positions. The eleven named omissions remain above their field's published maximum at 691. Accordingly, the broader failure limits *splitting alone*; a conjunction that first applies the root/width screens and then the splitting test still fits these recorded labels. Its historical use and motivation remain unverified. Treating the extra positions as original rejections would also require unavailable visitation evidence. The table gives every strict-691 counterexample.

| Seed position | Category | $S_{691}$ | Pub. maximum | Difference |
|---|---|---|---|---|
| p192-338 | O | 58 | 60 | -2 |
| p192-774 | O | 60 | 60 | 0 |
| p192-1006 | W | 56 | 60 | -4 |
| p192-1256 | O | 58 | 60 | -2 |
| p192-1477 | O | 59 | 60 | -1 |
| p192-1914 | O | 58 | 60 | -2 |
| p192-2678 | W | 55 | 60 | -5 |
| p192-2771 | W | 44 | 60 | -16 |
| p192-4254 | O | 57 | 60 | -3 |
| p192-5242 | W | 58 | 60 | -2 |
| p192-5674 | W | 52 | 60 | -8 |
| p192-5701 | W | 54 | 60 | -6 |
| p192-5964 | W | 56 | 60 | -4 |
| p192-5976 | O | 51 | 60 | -9 |
| p224-2328 | W | 50 | 50 | 0 |
| p239-155 | O | 56 | 57 | -1 |
| p239-765 | O | 56 | 57 | -1 |
| p239-1670 | O | 55 | 57 | -2 |
| p239-2228 | O | 56 | 57 | -1 |
| p239-2311 | O | 54 | 57 | -3 |
| p256-276 | O | 56 | 57 | -1 |
| p256-1383 | O | 56 | 57 | -1 |

Category W denotes order-width-only; O denotes other-root-only. The difference is the candidate's score minus the largest published score in its field.

## 4 Branches, the base point and coefficient

All 82 recorded branches remain explicit. Within each seed position, their discriminants are verified equal before selecting a representative for the 69-position comparison: canonical when present, otherwise the sole other branch. Isomorphic P-224 coefficient signs have equal order. A quadratic twist over the other fields has trace $-t$, and $(-t)^2 - 4p = \Delta$. Thus this score cannot distinguish those twist branches.

For $a = -3$, every on-curve point satisfies $b \equiv y^2 - x^3 + 3x \pmod{p}$. Hence $x = 0$ gives $y^2 = b$, while $x = 1$ gives $y^2 = b - 2$. These identities explain simple relations for assigned smallest-$x$ points. The earlier finite G–b study independently checked 100548 trait cells, found no separator among 19152 single-envelope tests, and no match in 880 fixed public-scalar checks [6]. It does not exhaust all mathematical functions, and the historical unpublished generators remain unknown. The present score uses only $p, n$ and adds no point attack.

## 5 Verification, provenance and interpretation

Source, inputs, commands and resource bounds were committed before actual cohort extraction at `1fe0d49974b2a3083772d003b69f4e75fe48aa4b`. The GMP producer checks nonsingularity, point equations, Hasse membership, uniqueness and $[n]P = O$, then enumerates quadratic residues. The separate PARI verifier uses proved `isprime`, `ellmul` and `kronecker`, and pairwise comparison differences. It reproves all 76 distinct field/order primes and reconstructs six files byte for byte: branch endpoints, characters, all seed/cutoff profiles, joint outcomes/margins, counterexamples and summary. The implementations share the GMP integer family and TSV I/O, which handles no curve or comparison arithmetic. No new production point count was needed.

All ten phases finish as expected: two arithmetic controls, producer, verifier and three corruption preparations exit zero; the three deliberately corrupted character/profile/cutoff copies are refused with native exit one. There are 54776 odd-prime branch characters, 338169 seed/cutoff profiles and 4901 joint cutoff outcomes. Every original branch, source file, phase and corrupted copy is retained in the local raw archive. No S3 upload or remote custody receipt is claimed. Existing-results analysis uses the repository's documented local development exception; no random-experiment gate is waived.

The finding is a restriction on the empirical criterion's scope. It does not establish the original decision program, either weak-publication or weak-omission hypothesis, or a designer's motive. Fewer strict split primes has no demonstrated ECDLP-hardness ordering on these inputs. Isogeny-based random-reducibility results have additional same-field/order/endomorphism assumptions and do not turn this count into an attack [4]. The requested 10000 accepted random prime-order curves per field and published percentiles remain unrun. Already examined large fields are not unseen holdouts. A finite test family cannot measure every mathematical difference.

## References


[1] asanso. Public strict splitting-count observation, 2026. Pinned provenance in the earlier strict-count study, `research/nist_skip_criterion_20261025/strict_elkies_20261029`.

[2] Prior lab analysis. November-1997 X9.62 acceptance-space replay, 24 October 2026. Source revision `351c9c021c7be044ade54dc7e352b33d3a772244`, `research/nist_seed_1997_acceptance_20261024/analysis.json`.

[3] I. Shparlinski and A. Sutherland. On the distribution of Atkin and Elkies primes. `https://arxiv.org/abs/1112.3390`, version 2.

[4] D. Jao, S. Miller and R. Venkatesan. Expander graphs based on GRH with an application to elliptic curve cryptography. `https://arxiv.org/abs/0811.0647`, version 2.

[5] PARI Group. PARI/GP documentation: `isprime`, `kronecker`, `ellinit`, `ellmul`. `https://pari.math.u-bordeaux.fr/dochtml/html/`.

[6] Prior lab study. Generator–coefficient comparison: all 21 curves, 29 October 2026. `research/nist_skip_criterion_20261025/generator_b_execution_20261029/REPORT.md`.

# A All 69 selected seed-position endpoints

P: published; N: named unpublished omission; W: order-width-only; O: other-root-only. These are source classification categories, not proven historical program dispositions. $R$ is retained separately. Representative choice and both branch endpoints are explicit in the raw 82-branch table. Full profiles include all cutoffs, not only these endpoints.

| Seed position | Cat. | $S_{691}$ | $R_{691}$ | $S_{5000}$ |
|---|---|---|---|---|
| p192-139 | P | 60 | 3 | 335 |
| p192-338 | O | 58 | 2 | 323 |
| p192-774 | O | 60 | 4 | 354 |
| p192-1006 | W | 56 | 2 | 307 |
| p192-1256 | O | 58 | 1 | 329 |
| p192-1477 | O | 59 | 1 | 347 |
| p192-1914 | O | 58 | 1 | 332 |
| p192-2034 | O | 64 | 0 | 361 |
| p192-2198 | N | 61 | 0 | 339 |
| p192-2524 | O | 64 | 1 | 331 |
| p192-2678 | W | 55 | 4 | 325 |
| p192-2771 | W | 44 | 2 | 314 |
| p192-2832 | O | 64 | 1 | 333 |
| p192-3087 | O | 71 | 3 | 351 |
| p192-3506 | O | 66 | 2 | 347 |
| p192-4254 | O | 57 | 4 | 323 |
| p192-4455 | O | 63 | 1 | 342 |
| p192-4492 | O | 64 | 0 | 325 |
| p192-5228 | P | 57 | 1 | 317 |
| p192-5242 | W | 58 | 5 | 334 |
| p192-5405 | N | 62 | 0 | 313 |
| p192-5674 | W | 52 | 2 | 311 |
| p192-5701 | W | 54 | 3 | 327 |
| p192-5964 | W | 56 | 3 | 341 |
| p192-5976 | O | 51 | 1 | 315 |
| p192-6013 | P | 58 | 1 | 303 |
| p224-77 | W | 55 | 1 | 334 |
| p224-777 | N | 53 | 4 | 318 |
| p224-1485 | N | 56 | 3 | 342 |
| p224-2232 | W | 64 | 2 | 319 |
| p224-2328 | W | 50 | 1 | 314 |
| p224-2938 | W | 54 | 0 | 329 |
| p224-2954 | W | 59 | 3 | 319 |
| p224-3191 | W | 61 | 2 | 339 |
| p224-3199 | W | 65 | 4 | 340 |
| p224-5736 | N | 59 | 2 | 325 |
| p224-5861 | W | 66 | 1 | 337 |
| p224-7026 | P | 50 | 3 | 299 |
| p239-21 | O | 68 | 2 | 326 |
| p239-36 | N | 71 | 2 | 347 |
| p239-155 | O | 56 | 1 | 309 |
| p239-765 | O | 56 | 1 | 328 |
| p239-1280 | O | 60 | 2 | 333 |
| p239-1562 | N | 67 | 3 | 337 |
| p239-1670 | O | 55 | 1 | 321 |
| p239-1769 | O | 68 | 1 | 325 |
| p239-2185 | P | 42 | 1 | 315 |
| p239-2228 | O | 56 | 4 | 346 |
| p239-2285 | O | 65 | 3 | 350 |
| p239-2311 | O | 54 | 1 | 327 |
| p239-2320 | P | 57 | 2 | 320 |
| p239-2491 | N | 63 | 2 | 352 |
| p239-2682 | N | 61 | 4 | 322 |
| p239-2863 | N | 66 | 1 | 329 |
| p239-3048 | P | 57 | 1 | 336 |
| p256-81 | O | 62 | 3 | 320 |
| p256-276 | O | 56 | 5 | 318 |
| p256-876 | N | 58 | 2 | 318 |
| p256-1049 | O | 67 | 3 | 336 |
| p256-1383 | O | 56 | 2 | 307 |
| p256-1842 | P | 57 | 2 | 328 |
| p384-592 | O | 58 | 1 | 320 |
| p384-1432 | O | 60 | 2 | 326 |
| p384-1462 | O | 56 | 1 | 332 |
| p384-1497 | W | 60 | 4 | 319 |
| p384-1626 | W | 60 | 1 | 308 |
| p384-1796 | O | 58 | 1 | 335 |
| p384-2688 | P | 54 | 0 | 324 |
| p521-1136 | P | 52 | 1 | 339 |